Hannah E. Holland-Moritz, Ph.D

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Hannah Holland-Moritz investigates the biotic and abiotic conditions which influence microbial community assembly and the molecular mechanisms behind community functional traits. Her work focuses on microbial communities in soils and plant-soil feedbacks with particular focus on arctic and boreal ecosystems. She uses molecular tools, bioinformatics, machine learning, and data science to identify microbial traits of interest and incorporate those traits in predictive biogeochemical models at the landscape to global scale.

EDUCATION

University of Colorado, Boulder	Ecology & Evolutionary Biology *Advisor: Dr. Noah Fierer*	Ph.D.	2021
University of California, Davis	Biochemistry & Molecular Biology with Honors *Minor: Quantitative Biology & Bioinformatics*	B.S.	2014

ACADEMIC POSITIONS

2023 – present	**Microbiome Research Scientist**, Department of Natural Resources and the Environment *University of New Hampshire, Durham NH, USA*
2021 – 2022	**Postdoctoral Researcher**, Department of Natural Resources and the Environment *University of New Hampshire, Durham NH, USA*
2015 – 2021	**Graduate Research Assistant**, Cooperative Institute for Environmental Sciences *University of Colorado, Boulder CO, USA*
Jun. 2020 – Aug. 2020	**Graduate Research Assistant**, Land Model Working Group *National Center for Atmospheric Research (NCAR), Boulder CO, USA*
2014 – 2015	**Junior Specialist Researcher**, UC Davis Genome Center *University of California, Davis, Davis CA, USA*
2011 – 2014	**Undergraduate Research Assistant**, UC Davis Genome Center *University of California, Davis, Davis CA, USA*

Affiliate Positions

2022-2023	**Affiliate Faculty,** Department of Forest and Rangeland Stewardship, *Colorado State University, Fort Collins, CO, USA*

PUBLICATIONS

*(Peer-Reviewed Total = 16, First Author = 3; * indicates co-first author, † indicates graduate mentee)*

Forthcoming (papers submitted, in review, and in revision; manuscript available upon request)

1. †Bradbury, E., Holland-Moritz, H., Gill, A., Havrilla, C. Plant and soil microbial composition legacies following indaziflam (Rejuvra™) herbicide treatment, 2024, *Submitted*
2. †Schaefer, S., Montano-Lopez, F., **Holland-Moritz, H,** Hicks Pries, C., Ernakovich, JE, 2024, Characterization of Rhizosphere and Mycorrhizal Communities of Common Tundra Plants, S*ubmitted*
3. Stuart, J., †Key, R., McDaniel, S., Hoffert, M., Fierer, N., **Holland-Moritz, H.**, Mack, M., 2024, Tundra moss transplants show host species-specific bacterial communities and independent responses of associated N_2 fixation rates to environmental change, *In revision*
4. †Kimmell, L., **Holland-Moritz, H.** Fagan, J.M.,Munson, S.M., Gehring, C.A., Havrilla, C.A., 2024, Differences in Microbiomes between intact and degraded sites suggest opportunities for dryland soil restoration, *Submitted*
5. †Freire Zapata, V., **Holland-Moritz, H.**, Cronin, D., Aroney, S., Smith, D., Wilson, R.M, Ernakovich, JE, Woodcroft, BJ, Emerge 2012 Field Team, EMERGE Coordinators, Rich, V., Sullivan, M., Stegen, J.C., Tfaily, M.M., 2024, Multi-omics unveils feature-level microbe-metabolite linkages driving permafrost greenhouse gas dynamics., *In revision*
6. **Holland-Moritz, H.**, Vanni, C., Fernandez-Guerra, A., Bissett, A., Fierer, N., *BioRxiv (Resubmitted)*, An ecological perspective on microbial genes of unknown function in soil https://doi.org/10.1101/2021.12.02.470747

Peer Reviewed

7. Slate, M., Antoninka, A., Bailey, L., Berdugo Moreno, M.B., Callaghan, D., Cárdenas, M., Chmielewski, M., Fenton, N., **Holland-Moritz, H.,** Hopkins, S., Jean, M., Kraichak, B. E., Lindo, Z.; Merced, A., Oke, T., Stanton, D., Stuart, J., Tucker, D.,; Coe, K. K., *New Phytologist*, 2024 Impact of changing climate on bryophyte contributions to terrestrial water, carbon, and nitrogen cycles
8. Rocci KS, Cleveland CC, Eastman BA, Georgiou K, Grandy AS, Hartman MD, Hauser E, **Holland-Moritz H**, Kyker-Snowman E, Pierson D, Reich PB. Aligning theoretical and empirical representations of soil carbon-to-nitrogen stoichiometry with process-based terrestrial biogeochemistry models. 2024, *Soil Biology and Biochemistry.*
9. †Texeira, C. D. S., **Holland-Moritz, H.,** Bayer, C. Eichelberger Granada, C., Sausen, T. S., Tonial, F. Petry, C. Frey, S. D. 2024 Land management of formerly subtropical Atlantic Forest reduces soil carbon stocks and alters microbial community structure and function, *Applied Soil Ecology*
10. Jay KR, Wieder WR, Swenson SC, Knowles JF, Elmendorf SC, **Holland-Moritz H,** Suding KN. Topographic heterogeneity and aspect moderate exposure to climate change across an alpine tundra hillslope. 2023, *Journal of Geophysical Research: Biogeosciences*.
11. Fierer, N., **Holland-Moritz, H.**, Alexiev, A., Batther, H., Dragone, N. B., Friar, L., Gebert, M. J., Gering, S., Henley, J. B., Jech, S., Kibby, E. M., Melie, T., Patterson, W. B., Peterson, E., Schutz, K., Stallard-Olivera, E., Sterrett, J., Walsh, C., and Mansfeldt, C., 2022, A metagenomic investigation of spatial and temporal changes in sewage microbiomes across a university campus *mSystems*

12. †Fan, K., **H. Holland-Moritz**, C. Walsh, X. Guo, D. Wang, Y. Bai, Y. Zhu, N. Fierer, H. Chu. 2022. Identification of rhizosphere microbes that actively consume plant-derived carbon. *Soil Biology & Biochemistry*.

13. Vincent, K., **Holland-Moritz, H.**, Solon, A.J., Gendron, E.M.S., Schmidt, S.K., 2022. Crossing Treeline: Bacterioplankton communities of alpine and subalpine Rocky Mountain Lakes *Frontiers in Microbiology*

14. Dragone, N. B., Henley, J. **Holland-Moritz, H.**, Diaz, M., Hogg, I., Lyons, W.B., Wall, D. Adams, B., Fierer, N., 2021, Elevational constraints on the composition and genomic attributes of soil microbial communities in the Transantarctic Mountains. *mSystems*

15. Stuart, J., **Holland-Moritz, H.E.**, Jean, M., Miller, S.N., Ponciano, J.M., McDaniel, S.F., Mack, M.C. 2021. The relationship of C and N stable isotopes to high latitude moss-associated N_2-fixation. *Oecologia*

16. **Holland-Moritz, H.**, Stuart, J., Lewis, L. R., Miller, S., Mack, M., Ponciano, J.M., McDaniel, S., and Fierer, N. 2021. The bacterial communities of Alaskan mosses and their contributions to N_2-fixation. *Microbiome*

17. Stuart, J., **Holland-Moritz, H.,** Lewis, L., Jean, M., Miller, S., McDaniel, S., Fierer, N., Ponciano, J. M., and Mack, M., 2020. Host identity as a driver of moss-associated N_2-fixation rates in Alaska. *Ecosystems*

18. Jean, M., **Holland-Moritz, H.,** Melvin, A., Johnstone, J., and Mack, M. 2020. Experimental assessment of tree canopy and leaf litter controls on the microbiome and nitrogen fixation rates of two boreal mosses. *New Phytologist*

19. Carini, P., Delgado-Baquerizo, M., Hinckley, E. S., **Holland-Moritz, H.** Brewer, T. E., Rue, G., Vanderburgh, C., McKnight, D., and Fierer, N. 2020. Effects of Spatial Variability and Relic DNA Removal on the Detection of Temporal Dynamics in Soil Microbial Communities. *mBio.*

20. Ward, L. M., Cardona, T., and **Holland-Moritz, H.** 2019. Evolutionary Implications of Anoxygenic Phototrophy in the Bacterial Phylum Candidatus Eremiobacterota (WPS-2). *Frontiers in Microbiology*

21. **Holland-Moritz, H.**, J. Stuart, L. R. Lewis, S. Miller, M. C. Mack, S. F. McDaniel, and N. Fierer. 2018. Novel bacterial lineages associated with boreal moss species. *Environmental Microbiology*

22. Gunawardana, M., S. Chang, A. Jimenez, D. Holland-Moritz, **H. Holland-Moritz**, T. P. La Val, C. Lund, M. Mullen, J. Olsen, T. a Sztain, J. Yoo, J. a Moss, and M. M. Baum. 2014. Isolation of PCR quality microbial community DNA from heavily contaminated environments. *Journal of microbiological methods*.

Non-Peer Reviewed

1. **Holland-Moritz, H. E.**, D. R. Bevans, J. M. Lang, A. E. Darling, J. A. Eisen, and D. A. Coil. 2013. Draft genome sequence of *Leucobacter sp.* strain UCD-THU (phylum *Actinobacteria*). *Genome announcements 1:e00325--13.*

2. **Holland-Moritz, H. E.**, D. A. Coil, J. H. Badger, G. I. Dmitrov, H. Khouri, N. L. Ward, F. T. Robb, and J. A. Eisen. 2014. Draft genome sequence of the pyridinediol-fermenting bacterium *Synergistes jonesii* 78-1. *Genome announcements* 2:e00833–14.

GRANTS, FELLOWSHIPS, AND FUNDING

(research grants totaling $210,753)

Funded (indicates PI, ** indicates co-PI)*

Grant	*Ernakovich, JE, **Hicks Pries, C, Holland-Moritz, H. Monano Lopez, Wieder, W. Grandy "FICUS: A high-resolution view of the plant–microbe–mineral interactions affecting C-cycling in thawed permafrost soils", *Integrating Collaborations for User Science (FICUS) Award, DOE* • Contributed to study design, outline, and core writing of research background	2023	$204,753
Grant	*Holland-Moritz, H., "Sandbox Succession: Functional succession of microbial communities in a climate context" *Department Graduate Research Grant, CU Boulder*	2019	$2500
Grant	*Holland-Moritz, H. "Sandbox Succession: Functional Succession of Microbial Pioneers in a Climate Context" *Beverly Sears Graduate Student Grant, CU Boulder*	2019	$1000
Grant	*Holland-Moritz, H., Dept. of Ecology and Evolutionary Biology (EBIO) Graduate Research Grant	2018	$2500
Travel Grant	Departmental Graduate Travel Grant, *CU Boulder*	2018	$300
Travel Grant	Graduate School Domestic Travel Grant, *CU Boulder*	2017	$500
Scholarship	Wood's Hole Marine Biological Laboratory Summer Fellowship, *Wood's Hole Marine Biological Laboratory*	2016	$8751

Unfunded/Pending (indicates PI, ** indicates co-PI)*

Grant	*Ernakovich, JE, **Holland-Moritz, H, "Resolving How Soil Organic Matter Degrading-Functional Groups Contribute to Priming in the Arctic Rhizosphere", *Environmental Systems Science Program, DOE* • Led conceptualization, development, writing of proposal related to current Arctic Systems Science Funding Grant (NSF #2031253); [Listed as co-PI because only current faculty can serve as PI at UNH]	pending	$999,876
Grant	*Frey, S., **Ernakovich, JE, **Grandy, S., **Holland-Moritz, H., "Microbial trait and genomic drivers of soil organic matter formation and stabilization", *Hatch NH Program New Hampshire Agricultural Experimental Station, UNH* • Contributed to development and conceptualization of 'omics-carbon modeling integration	pending	$138,198
Grant	*Kuhn, M., **Holland-Moritz, H., **Arndt, K., **Virkkala, A., Improving our understanding of the northern methane sink through better representation of alpine tundra ecosystems, *Spark Climate Solutions (Non-profit)* • Contributed to development and design, and writing of microbial	2024	$296,655

functional analyses

Grant	*Holland-Moritz, Advisors: Ernakovich, JE and Brodie, E. “The role of microbial dispersal in northern wetland carbon release: modeling genomes-to-ecosystems”, *Postdoctoral Research Fellowship in Biology, NSF*	2021
Grant	*Frey, S. et al. Microbiome Science Enabled Through Environmental Genomic Approaches (NH-MEGA) *RII-Track I NSF EPSCOR program* • Contributed to conceptualization, development, and writing with focus on data-model integration, microbiome response to disturbance, and broader impacts related to computational training of future STEM workforce	2020
Grant	*Fierer, N, **McDaniel, S. **Mack, M., Holland-Moritz, H. Assembly and function of the nitrogen-fixing microbiome of a keystone moss species, *Community Sequencing Program, DOE* • Aided in methods development and figure generation, contributed to hypothesis writing, goals and sequencing plan	2016
Grant	*Holland-Moritz, H., NSF-GRFP Honorable Mention *- The Diversity and Activity of Moss Associated Nitrogen-Fixing Bacteria in Boreal Ecosystems* *- Sandbox Succession: Phylogenetics and Function of Microbial Pioneers in Climatic Context* *- Exploring Disturbance in Microbial Systems through Winogradsky Model Communities*	2017 2016 2015

PRESENTATIONS AND INVITED TALKS

Seminars and Invited Talks (indicates equal first or senior author contribution,)*

2024 **Holland-Moritz, H.** Using microbial traits to connect to ecosystem processes through models, *BRiDGE2Science, Stockbridge School of Agriculture*, University of Massachusetts, Amherst, Amherst MA.

2024 **Holland-Moritz, H.,** Practicing Finding and Using Your Power: A manual for early career scientists, *BRiDGE2Impacts, Stockbridge School of Agriculture*, University of Massachusetts, Amherst, Amherst MA.

2023 **Holland-Moritz, H***, Dylan Cronin*, Derek Smith*, Samuel Aroney*, Carmody McCalley, Kelly Wrighton, Mikayla Borton, Suzanne Hodgkins, Rory Flynn, Yueh-Fen Li, Isogenie Field and Analytic Teams, EMERGE Institute Coordinators, Virginia Rich*, Sarah C. Bagby*, Ben Woodcroft*, Gene Tyson*, Jessica Ernakovich*. Functional Ecology and Community Assembly Buffer Climate Feedbacks Across a Permafrost Thaw Gradient: A Genome-Resolved, Seven Year Study, *American Geophysical Union Annual Meeting*, San Francisco, CA.

2023 Ernakovich, J.G., Bobyock, J., Doherty, S.J., **Holland-Moritz, H.**, O'Brien, J.M., Schaefer, S., Schlerman, E. Microbial dynamics following permafrost thaw: how does disturbance intensity and time affect microbiome structure and biogeochemical function? *American Geophysical Union Annual Meeting*, San Francisco, CA.

2023 Rich, V.I., Li, Z., **Holland-Moritz, H.,** Varner, R.K., Saleska, S.R., Riley, W.J., Wehr, R., Tang, J., Brodie, E., Chanton, J., Wrighton, K., Sullivan, M.B., Ferrière, R., Tfaily, M.M, Bagby, S., Tyson, G.W., Woodcroft, B.J., Ernakovich, J.G., Frolking, S.E., Zayed, A.A., Palace, M.W., Wu, Q, and The EMERGE Biology Integration Institute. An ecosystem of models to model an ecosystem. *American Geophysical Union Annual Meeting*, San Francisco, CA.

2023 **Holland-Moritz, H.**, López, F.M., Schaefer, S., Schlerman, E., Bakke, P., Wieder, W., Grandy, A.S., Hicks Pries, C., Ernakovich, J.G. How systems thinking can help narrow uncertainty in priming

outcomes in permafrost soils. *ASA, CSSA, SSSA International Annual Meeting*, St. Louis, MO

2022 **Holland-Moritz, H.**, Vanni, C., Fernandez-Guerra, A., Bissett, A., Fierer, N., An ecological perspective on microbial genes of unknown function in soils, *Brodie group meeting*, Lawrence Berkeley National Lab

2022 **Holland-Moritz, H.** Phylogenetic and environmental controls on the structure of boreal moss microbiomes, in "From desert to the Arctic: ecological significance of bryophytes across systems", ESA, Montreal, Canada (Lightning talk and invited panelist)

2022 **Holland-Moritz, H.** Dylan Cronin*, Derek Smith*, Samuel Aroney*, Carmody McCalley, Kelly Wrighton, Mikayla Borton, Suzanne Hodgkins, Rory Flynn, Yueh-Fen Li, Isogenie Field and Analytic Teams, EMERGE Institute Coordinators, Virginia Rich*, Sarah C. Bagby*, Ben Woodcroft*, Gene Tyson*, Jessica Ernakovich* Assembly of microbes in thawing permafrost, *Extreme Microbiome seminar series*, virtual

2021 **Holland-Moritz, H.,** Microbes, mosses, and metagenomes, *Ecology and Evolutionary Biology group seminar*, Case Western University, virtual

2021 **Holland-Moritz, H.** Dylan Cronin*, Derek Smith*, Samuel Aroney*, Carmody McCalley, Kelly Wrighton, Mikayla Borton, Suzanne Hodgkins, Rory Flynn, Yueh-Fen Li, Isogenie Field and Analytic Teams, EMERGE Institute Coordinators, Virginia Rich*, Sarah C. Bagby*, Ben Woodcroft*, Gene Tyson*, Jessica Ernakovich*Microbial Assembly in Thawing Permafrost, Regional Conference on Permafrost, Boulder, CO USA/Virtual (invited talk)

2020 **Holland-Moritz, H.,** Microbes, mosses, and metagenomes, *Departmental Colloquium*, University of Colorado, Boulder, Boulder CO

2020, 2018, 2017 **Holland-Moritz, H.** Research Experience for Community College Students (RECCS) Program, Q&A panelist and invited talk "How to communicate your research."

2014 "rRNA Sequence Analysis in QIIME." UC Davis Bioinformatics Club, invited talk

Contributed Talks (indicates equal first or senior author contribution, † indicates graduate student mentee, § indicates undergraduate mentee)*

2023 †Bratsman, S., Ernakovich, J.G., Fahnestock, M.F., Varner, R.K., McCalley, C.K., **Holland-Moritz, H.**, DeFelice, H., Rodriguez, Y.G., Ortega, M., Rodríguez, A.L., Wilcox, M., Atley, E.T. Microbial and Carbon Feedbacks During Thaw-Induced Transitions in Arctic Peatlands. *American Geophysical Union Annual Meeting*, San Francisco, CA.

2023 †Blais, N., Warren, N., Teixeira, C. dS., **Holland-Moritz, H.**, Smith, R., Castillo, B., §Krein, M., Ernakovich, J.G. The Effects of Grass Legume Bi-culture Defoliation on Soil Microbial Communities and Physico-chemistry. *ASA, CSSA, SSSA International Annual Meeting*, St. Louis, MO

2022 *Aroney, S., *Smith, D., *Cronin, D.,***Holland-Moritz, H.**, Hodgkins, S., *Ernakovich, J., *Rich, V., *Bagby, S., *Woodcroft, B., *Tyson, G.. A fork in the metabolic road: mapping methane and carbon dioxide emissions from a microbial community in a thawing permafrost ecosystem. *Australian Microbial Ecology*, Melbourne, Australia.

2022 ***Holland-Moritz, H.**, *Cronin, D., *Aroney, S., *Smith, D., Hodgkins, S., *Woodcroft, B., *Tyson, G., *Rich, V., *Bagby, S., *Ernakovich, J. Biological and environmental drivers of microbial community assembly in a thawing permafrost ecosystem, *Ecological Society of America*, Montreal, Canada

2021 ***Holland-Moritz, H.**, *Cronin, D. EMERGE Institute Field Team, *Rich, V.I., *Ernakovich, J.G. Microbial Assembly in Thawing Permafrost, *American Geophysical Union*, New Orleans, LA USA/Virtual

2021 ***Holland-Moritz, H.**, *Cronin, D. EMERGE Institute Field Team, *Rich, V.I., *Ernakovich, J.G. . Assembly of microbial communities in thawing permafrost. *Regional Conference on Permafrost*, Boulder, CO (virtual).

2021 †O'Brien, J., †Blais, N., §Bobyock, J., Bernhardt, L., **Holland-Moritz, H.**, Douglas, T., Sullivan, T., Bailey, V., Barbato, R., Ernakovich, J.G.. Investigating microbial dormancy within the permafrost microbiome. *Regional Conference on Permafrost*, Boulder, CO (virtual).

2020 **Holland-Moritz, H.**, Henley J., Fierer, N., Primary succession of microbial communities in terrestrial environments: an experimental cross-site investigation, *Ecological Society of America General Meeting*, Salt Lake City, UT, USA/Virtual Meeting

2019 **Holland-Moritz, H.**, Stuart J.M, Lewis, L., Miller, S., Mack, M., Ponciano, J.M., McDaniel, S., Fierer N, Phylogenetic and Environmental Controls on N_2-fixation in the boreal moss microbiome, *Front Range Microbiome Symposium*, Fort Collins, CO, USA. (Contributed Talk and poster)

2017 **Holland-Moritz, H.**, Stuart J.M, Lewis, L., Miller, S., Mack, M., Ponciano, J.M., McDaniel, S., Fierer. Taxonomic and Functional Attributes of the Boreal Moss Microbiome, *Future Arctic Conference on Bryophytes and Lichens*, Quebec, Canada

2017 **Holland-Moritz, H.**, Stuart J.M, Lewis, L., Miller, S., Mack, M., Ponciano, J.M., McDaniel, S., Fierer, Taxonomic and Functional Attributes of the Boreal Moss Microbiome, *Guild of Rocky Mountain Ecologists and Evolutionary Biologists*, Nederland, CO, USA

Poster Presentations († indicates graduate student mentee, § indicates undergraduate mentee)

2023 Rodríguez, A.L., Rodriguez, Y.G., Ortega, M., Wilcox, M., Atley, E.T., †Bratsman, S., **Holland-Moritz, H.,** Fahnestock, M.F., Varner, R.K., Ernakovich, J.G. The Wettening of Arctic Peatlands: A Microbial Perspective on Evolving Arctic Landscapes. *American Geophysical Union Annual Meeting*, San Francisco, CA. [EMERGE REU]

2023 **Holland-Moritz, H.,** Hicks Pries, C., Wieder, W., Grandy, A. S., Ernakovich, J. "Strategies for translating microbial 'omics into function in models of Arctic priming" EMERGE Institute Workshop, Tuscon AZ, USA

2022 §Biletzkov, T., §Scalora-Riley, T., §Stoddard, P., Cory, A., Bosman, S., **Holland-Moritz, H.**, Varner, R., Fahnestock, F., Chanton, J. "Presence of Phenolic Compounds in Porewater at Stordalen Mire", *American Geophysical Union Annual Meeting*, San Francisco, CA. [EMERGE REU]

2022 §Stoddard, P., **Holland-Moritz, H.**, Ellenbogen, J., §Scalora-Riley, T., §Biletzkov, T., Fahnestock, F., Varner, R., Wrighton, K., "Redox profiles across thawing permafrost in Stordalen Mire" *American Geophysical Union Annual Meeting*, San Francisco, CA. [EMERGE REU]

2022 §Scalora-Riley, T., §Stoddard, P., §Biletzkov, T., Kuhn, MA., Bernhardt, L, Alexander, N., **Holland-Moritz, H.,** Fahnestock, M.F., Varner, R.K, Ernakovich, JE. "Dispering microbial communities increases variability in methane fluxes in arctic Sweden" *American Geophysical Union Annual Meeting*, San Francisco, CA. [EMERGE REU]

2018 **Holland-Moritz, HE**, Stuart, J, Lewis, LR, Ponciano, JM, McDaniel, S, Mack, M, Fierer, N

"Phylogenetic and Environmental Controls on the Boreal Moss Microbiome" Poster Presentation, 2018, *International Society for Microbial Ecology Meeting*, Leipzig, Germany.

2014 **Holland-Moritz, HE**, Lang, JM, Stachowicz, J, Eisen, JA "Exploring the biogeography of microbial communities on the surface of seagrasses." Poster Presentation: *20th Annual International Conference on Microbial Genomics* 2014 Lake Arrowhead, CA, USA.

2014 **Holland-Moritz, HE,** Lang, JM, Stachowicz, J, Eisen, JA "The Seagrass Microbiome" Poster Presentation at: University of California, Davis *Undergraduate Research Symposium* 2014. Davis, CA, USA.

TEACHING

Courses

Co-instructor – Graduate Course: Sewage Metagenomics 2020
- Pandemic-era hands-on research course focused on metagenomic analysis
- Co-developed readings to familiarize students with latest research and techniques in microbial 'omics analyses, and sewage microbial source tracking
- Prepared lessons in collaborative coding, informatics analysis, and data organization for students
- Helped students develop, write, and publish scientific article on course findings

Lab Instructor – Undergraduate Course: Principles of Ecology 2017
- Instructed two sections of ecology laboratory science (30 students total)
- Designed laboratory lecture material, assessments, and activities to train students in ecological research methods, reading and writing scientific literature, and basic ecological statistics
- Implemented group learning, and hands-on activities to teach students to analyze ecological data, write a research paper, and design and carry out research projects
- Assessed effectiveness of course material through use of pre- and post-surveys

Workshops, Guest-lectures, and Teaching Modules

(indicates workshop materials are freely available from my website)*

UNH Soil Ecology guest lecture and paper discussion activity: 2024
- Cross-listed graduate/undergraduate course for students from broad interdisciplinary backgrounds (20 students)
- Designed and led paper discussion on topic of utility of 'Omics tools in Soil Microbial Ecology

UNH-CRREL Bioinformatics Group – Introduction to Metagenomics Workshop Series 2021, 2022

Instructor – EMERGE Biology Integration Institute Summer Program 2021- 2024
- Two-week intensive early career-focused workshop on interdisciplinary ecosystem science)
- 120 participants over 4 years, ranging from undergraduate REU students to assistant professors
- Designed, developed, and taught three modules for the workshop: *Introduction to Community Assembly, Tools of Environmental Microbiology, *An ecosystem of models: Using Predictive Models Across Spatial and Temporal Scales

Lead instructor – *Using Make and Makefiles for automating analysis pipelines workshop 2020

- Designed and tailored workshop for the needs and tools of research ecologists
- Advertised, led and adapted workshop for virtual presentation during the Covid-19 pandemic

Lead instructor – *Collaboration and Version Control with git Workshop and Hackathon 2020

- Two-day workshop to teach graduate students how to use version control software and apply that knowledge to real-world collaborations
- Designed adaptable lesson plan, customizable to student learning goals

Lead instructor – Science and the Creative Process: Training Creative Thinking (Workshop) 2019, 2021

Co-instructor – *High-throughput sequence data workshop series 2018

- Co-Designed a pipeline and tutorial in use of high performance computing environments to analyze microbial amplicon sequence data with the dada2 pipeline
- Held series of workshops to train students and users through CU Boulder's Center for Microbial Exploration
- Manage pull-requests and train students to contribute to pipeline
- Published and maintained tutorial on github, integrated updates for handling sequencing data from the NovaSeq platform
- >1200 unique visitors, 531 unique clones, 11 stargazers, and 2 forks on github in last year

Pedagogical Training Training and Professional Development

Mentorship Academy – UNH 2022

Graduate course in Mentorship and Leadership in STEM 2017

- Course focused on best practices in academic mentorship

MENTORSHIP AND ADVISING

** indicates service on student's thesis committee*

Graduate Students

Rebecca Key (PhD, University of Florida)	2023 - present
Sam Bratsman (PhD)	2022 - present
Adeena Ahsan (MS) *	2022 - present
Alma Hernandez (MS)	2021 - present
Sean Schaefer (PhD)	2021 - present
Nate Blais (MS/PhD)	2021 - present
Viviana Freire Zapata	2021-present
Ember Bradbury (PhD, Colorado State University)	2022 - present
Cristhian Teixeira (PhD)	2022
Louisa Kimmell (MS, Colorado State University) *	2021-2023
Joy O'Brien (MS)	2021-2022

Undergraduates

Torin Scalora-Riley (Current SURF mentor)	Summer 2022 - present
Tallulah Biletzkov	Summer REU 2022

Parker Stoddard	Summer REU 2022
Mikayla Lopez	Summer 2021
Savanna Pierce	Summer 2019
Lady Grant	2018 - 2019
Maya Montoya-Pimolwatana	2017 - 2018
Alexander Kaaua	2015 - 2016
Ruth Lee	2014 - 2015
Briana Pompa-Hogan	2014 - 2015
High School Students	
Henna Hundal	Summer 2014

AWARDS

Dept. of Ecology and Evolutionary Biology Graduate Student Service Award	2019
Integrated Studies Honors Scholar	2011 - 2014
University of California Regents Scholar	2010 - 2014

RELEVANT SKILLS

Computational: Terminal/Bash (proficient), Git version control (proficient), R (proficient), Linux (proficient), use of high power computing systems (proficient), Python (familiar with), Perl (familiar with), C (familiar with), LaTeX (familiar with)

Bioinformatics and Statistics: Genome assembly, metagenomics (read-based and assembly), environmental amplicon-sequencing analysis, phylogenetic tree-building, gene annotation, plotting and visualization of biological data, hierarchical bayesian modeling, terrestrial climate modeling with Community Land Model (CLM), machine learning

Laboratory: cell culture, defined culture media design, PCR, amplicon library sequencing, DNA/RNA extraction, metagenomics, metatranscriptomics, genomic library preparation

Language (ILR Scale): English (native), French (full professional proficiency), German (limited working proficiency), Japanese (limited working proficiency), Arabic (elementary proficiency) [*Language learning is a personal hobby.*]

Other: Social Media (Twitter: @hhollandmoritz)

SELECTED SYNERGISTIC ACTIVITIES

1. *Educator*
    - Lead instructor in numerous data science skills workshops including workshops focused on version control, make, reproducible science, and best practices in scientific coding (2018-present)

- Created microbial analysis pipeline tutorial and led workshops in its use for CU Boulder Center for Microbial Exploration, and UNH's SoilBioME group. (2019-2022)
- Contributed multiple workshops on quantitative topics including R, multivariate analysis, and Bayesian statistics to EBIO Department's weekly Quantitative Discussion working group (2016-2019).

2. *Community Service*

- Created, organized and co-run SoilBioME writing cooperative – a collaborative peer-writing group of soil scientists and students (2022)
- EMERGE-BII (Biology Integration Institute) DEI committee. Led efforts to improve cross-institutional reporting structure, community agreement, and field safety (2022)
- Co-organized the departmental graduate student orientation. (2020, 2017)
- Served as the Graduate Student Representative to the Faculty. Led efforts to improve mentorship, financial stability, and accessibility for graduate students. (2017-2019)
- Colloquium committee member and travel coordinator for weekly colloquium series for CU Boulder's EBIO Department. (2015 - 2019)
- Graduate student representative on Departmental Climate and Graduate Degree Program Committees for departmental 7-year Academic Review and Planning Advisory Committee review. (2018)
- Graduate student representative to two faculty search committees. (2018)
- Organized and ran departmental graduate student writing cooperative. (Summer 2017)
- Drafted departmental Inclusion, Diversity, and Excellence in Academics (IDEA) plan, led initiative to increase diversity of graduate student applicant pool, and foster an inclusive graduate student environment as member of EBIO Diversity and Inclusion Working Group (2017-2021)

3. *Mentor*

- Research mentor to 7 graduate students (2 as a committee member), 10 undergraduates, and 1 high school student
- Served as a peer mentor for four graduate students in departmental mentoring program. (2016-2019)

4. *Science Communicator*

- Scientist guest speaker, Skype a Scientist, at 5 K-12 classrooms
- Designed, coordinated, and carried out community science research project focused on microbial succession with middle and high school educators across the USA. (2018 - 2019)
- Co-chair and participant of CU Boulder Evolution Outreach Committee. Annually organized continuing education "Teaching Controversial Science Topics" workshop for local middle and high school teachers, and Museum Family Day on evolution. (2015 – 2019)
- "Mysterious Microbes of Soil" Workshops and Webinar series. Helped organize, plan, and run workshops for K-12 teachers and gave public webinar presentations focused on microbial ecology. (2016)

5. *Reviewer*

- For journals including: Applied and Environmental Biology, Global Change Biology, PeerJ, American Journal of Botany, Ecosystems, Functional Ecology, Trends in Microbiology, Oecologia, Microbiome, ISMEJ, Journal of Ecology, New Phytologist, and Science of the Total Environment.

- Invited judge for undergraduate research presentations at the annual EBIO departmental Spring Symposium (2018).

FURTHER RELEVANT RESEARCH & WORK EXPERIENCE

Participant, NSF EMERGE-BII Summer Institute, Virtual 2021
- a two-week workshop on integrative ecosystem biology with courses in modeling, microbial processes, multi-omics integration, isotopic gas flux measurements, team science, and mentoring

Participant, Anvi'o workshop, University of California, Irvine 2017
- a multi-day workshop focused on the use of Anvi'o in 'omics analyses

Participant, MBL Microbial Diversity Course, Woods Hole, MA 2016
- a 6.5 week course dedicated to training students in laboratory and theoretical aspects of microbial diversity

Research Internship, Drs. Manjula Gunawardana and Marc Baum, Oak Crest Institute of Science 2009 - 2010
- Researched extraction methods of PCR-quality DNA from heavily hydrocarbon-polluted environmental samples (e.g. asphalt pools).

REFERENCES

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